

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

Bhaiichara on
TOP.

STRUCTURE OF ATOM

LECTURE-5

CHEMISTRY

VISHAL SIR

JKBose Wallah



Topics To Covered

- 1 DUAL NATURE OF MATTER AND RADIATION**
- 2 DE-BROGLIE RELATIONSHIP**
- 3 HEISENBERG'S UNCERTAINTY PRINCIPLE**

CHAPTERS IN CHEMISTRY (11th).

- 1) SOME BASIC CONCEPTS OF CHEMISTRY. (PC)
- 2) STR. OF ATOM. (PC)
- 3) Classification of Elements + Periodic Properties. (IOC)
- 4) Chemical BONDING. (IOC)
- 5) Thermodynamics (PC)
- 6) Equilibrium (PC)
- 7) Redox Reaction (PC)
- 8) Some Basic Principles and techniques of OG. (OG)
- 9) HYDROCARBONS. (OG)

Chemistry:

100/100

10

Q. We Need Two Teachers in Chemistry?

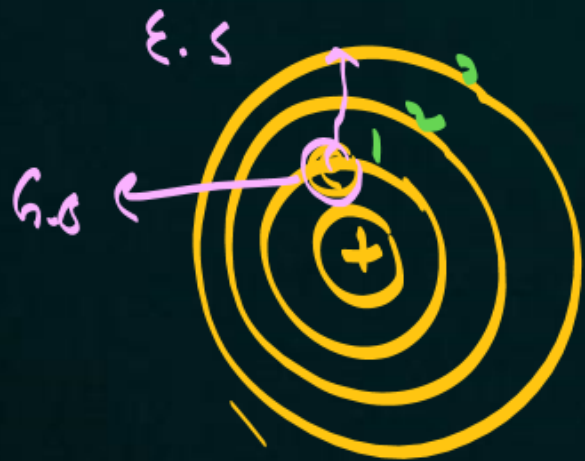
Why ———

→ Halwa ✓

REVISION

BOHR MODEL OF ATOM $\rightarrow$

- $n = 1, 2, 3, 4, 5 \dots$
K, L, M, N,



- Energy (For one e^- species)
 $H, H^+, Li^+, etc.$

$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J/atom.}$$

$$E_n = -13.6 \frac{Z^2}{n^2} = \text{eV/atom}$$

$$E_n = -1312 \frac{Z^2}{n^2} \text{ ———}$$

- Angular momentum

$$mve = n \frac{h}{2\pi}$$

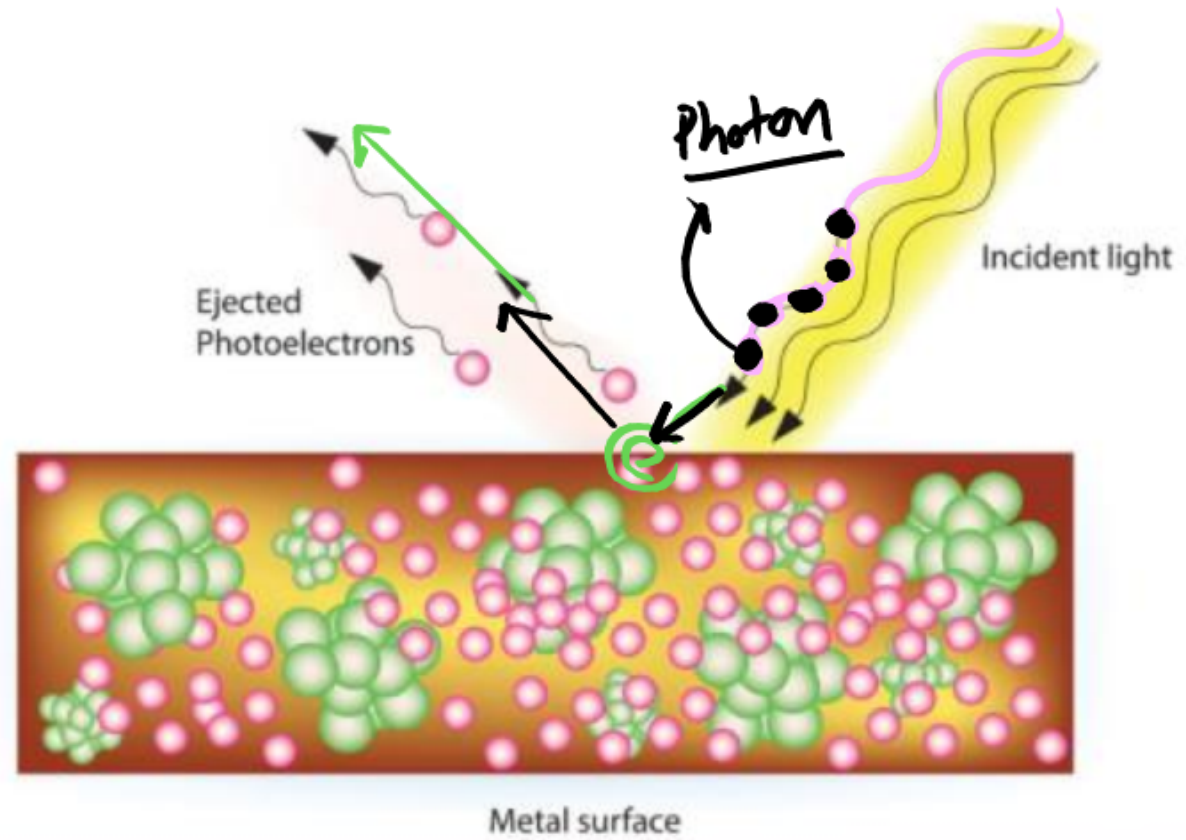
- Radius of

$$r_n = a_0 \frac{n^2}{Z}$$

$$a_0 = 0.529 \text{ \AA} \text{ or } 52.9 \text{ pm}$$

$$\Delta E = E_{\text{final state}} - E_{\text{G.S.}}$$

Photoelectric effect

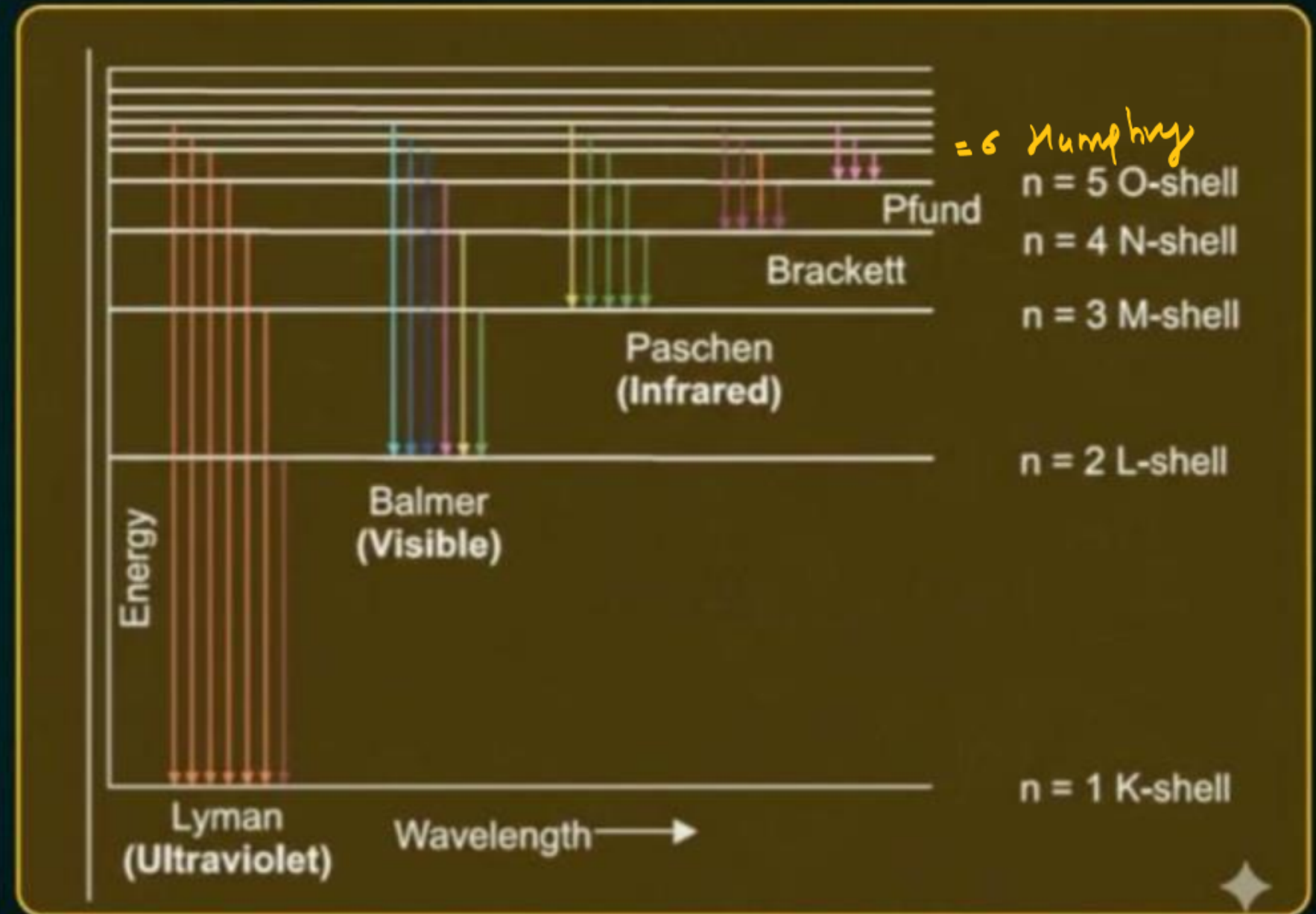


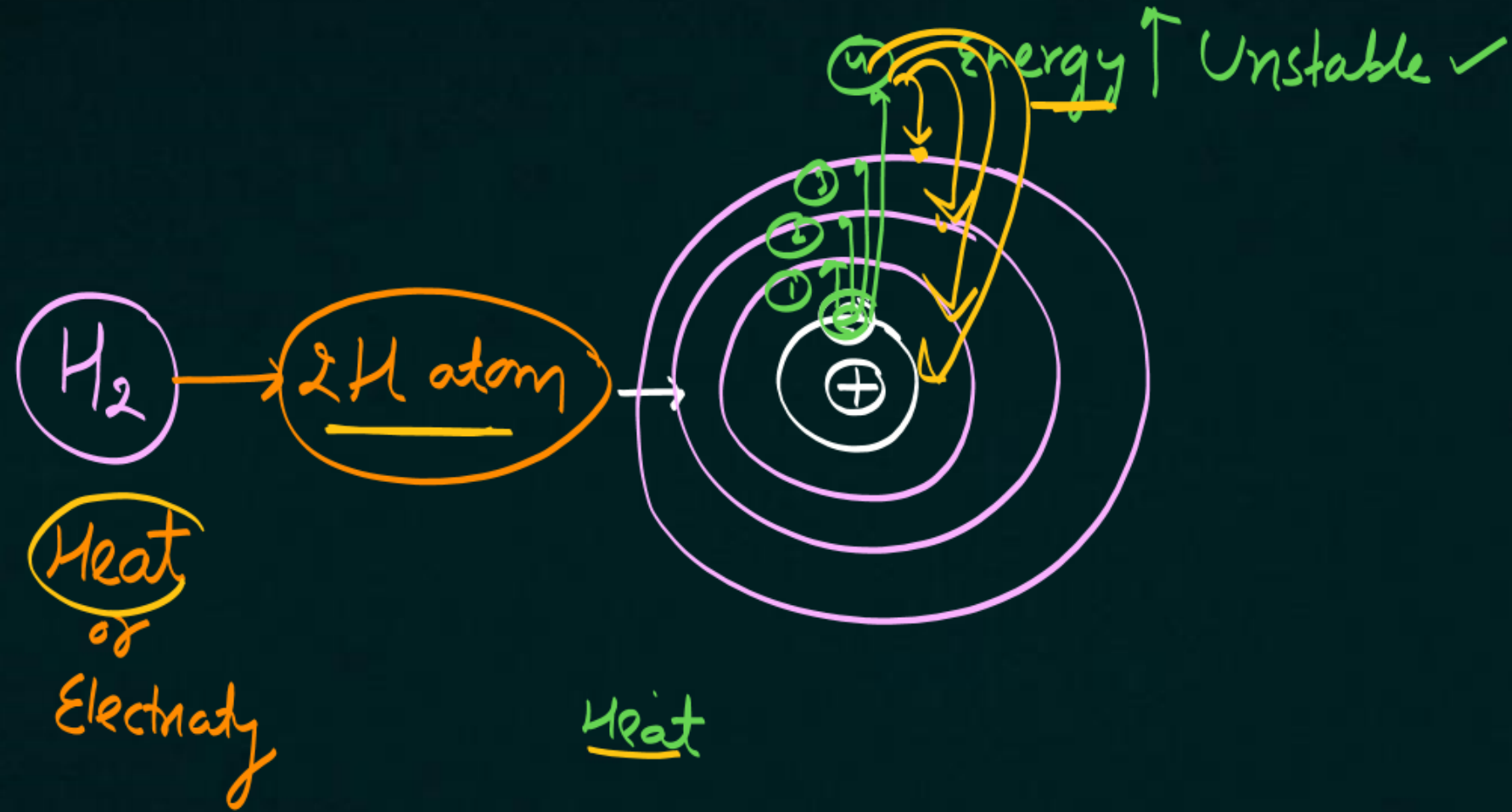
Line Spectrum of Hydrogen

(Not mentioned in Syllabus).



- Heating or electric discharge splits hydrogen molecules into atoms.
- Electrons absorb varying amounts of energy and move to higher levels (e.g., L, M, N, O).
- The excited state of electrons is short-lived.
- Electrons return to lower levels or the ground state in one or more jumps.





BOHR MODEL OF ATOM



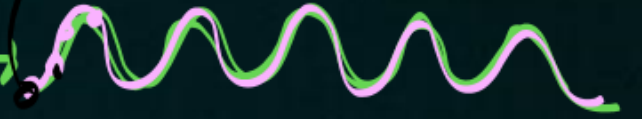
Limitations:

- (i) *Inability to explain line spectra of multi-electron atoms.*
- (ii) *Inability to explain splitting of lines in the magnetic field (Zeeman effect) and in the electric field (Stark effect).*
- (iii) *Inability to explain the three dimensional model of atom.*
- (iv) *Inability to explain the shapes of molecules.*
- (v) *Inability to explain de Broglie concept of dual character of matter and Heisenberg's uncertainty Principle.*

Dual nature of Radiation and matter.

We know that light show so many phenomena like reflections, refraction, photoelectric effect, Black body radiations. Process like reflection, refraction can be studied only if we consider light as wave. But processes like photoelectric effect or Black body radiation we have to consider light consists of a small packets of energy (Photon) i.e particle nature. i.e we can say that radiations have dual nature i.e wave as well as particle.

light consist of small packet of energy $\rightarrow$ Photon.

Wave $\rightarrow$ 

Particle nature $\rightarrow$ Small packets with energy:

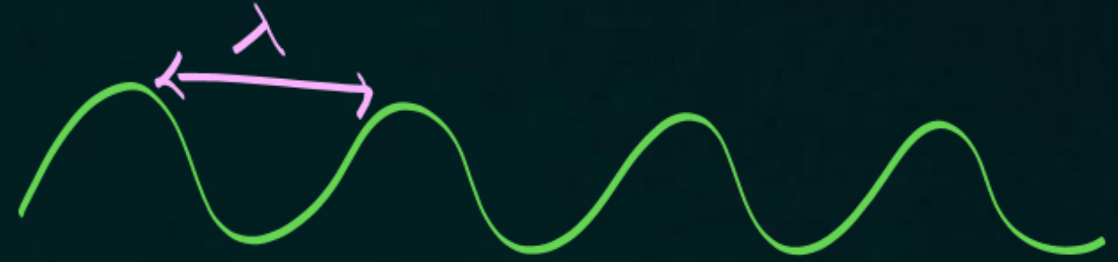
De broglie stated that like radiation, small microscopic particles like e^- , neutron, proton, pen , cluster etc also have dual nature i.e wave as well as particle.

And the wavelength of wave in which they travel can also be calculated as

$$\lambda = \frac{h}{mv}$$

h = planck's constant ($6.626 \times 10^{-34} \text{ Js}$)
 m = mass of particle
 v = velocity of the particle.

This relationship is known as De-broglie relationship



REVISION ko Aadat Bana Lo.

Thank
You